AI-Driven Intelligent Routing and Secure Autonomous Recovery in 6G-Integrated Disaster-Monitoring Sensor Networks

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Ph. D. Thesis in Computer Engineering

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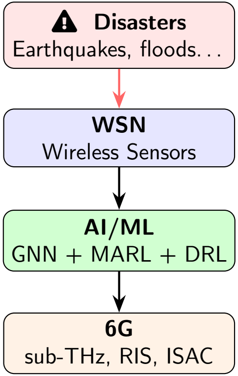
Outline

Context: Natural Disasters and Monitoring

- **7,348** major disasters (2000–2019)
  4.2 B people affected, 1.23 M deaths, \$2,970 B in losses [UNDRR 2020]
- In 2023: 398 disasters, 95,000 deaths, ∼\$380 B
- Climate change ⇒ +10.5% intensity, +30% frequency of extreme precipitation [IPCC AR6]
- **Sendai Framework 2015–2030**: strengthen early warning systems



Need

Real-time monitoring via Wireless Sensor Networks (WSNs) — reliable, sustainable, and secure.

WSNs: Potential and Constraints

WSN Strengths

- Autonomous multi-hop data collection
- Coverage of large geographical areas
- Post-disaster operation capability
- Low per-unit cost

Severe Constraints

- Non-rechargeable batteries ($\sim$0.5 J)
- Limited RAM (16–256 KB)
- Restricted bandwidth
- Physical vulnerability in hazardous zones

6G Enabling Technologies (horizon 2030)

sub-THz	**RIS**	**ISAC**
Multi-Gbit/s	Passive wavefront ctrl	Sensing + comm.

O-RAN: near-RT RIC $\leq$ 10 ms $\Rightarrow$ AI in the control loop

Three Fundamental Challenges

C1: Energy

- Routing = #1 factor for lifetime
- QMIX/QTRAN: $\sim$39K params/agent; QPSOFL: collapses at scale
- No topology-aware MARL policy fusion exists

C2: Carbon

- Embodied $\gg$ operational (10^6:1)
- No protocol considers full lifecycle
- No modulation by grid intensity

C3: Security

- Rogue policy injection
- Out-of-distribution hallucinations
- Binary mechanisms only (allow/block)

Research Questions and Objectives

Main Research Question

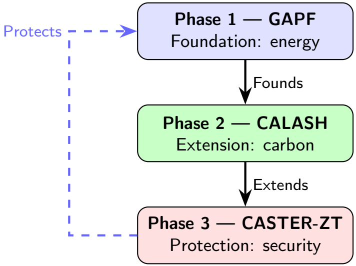
How to design a complete intelligent routing protocol stack — energy-efficient, carbon-sustainable, and secure — for AI-native 6G-integrated disaster-monitoring sensor networks?

	Specific Question	Specific Objective
Q1	Can a lightweight GNN encoder fuse complementary MARL policies to achieve a packet delivery ratio $\geq$ 98% with fewer than 2 000 trainable parameters on embedded platforms?	**SO1:** Develop GAPF, a GCN-based contextual algorithm selector that dynamically fuses QMIX and QTRAN experts with $<$ 2 000 parameters, achieving PDR $\geq$ 98% at 2–3.5$\times$ lower energy.
Q2	To what extent can a four-pillar carbon-aware protocol, combined with 6G physical-layer capabilities, reduce the life-cycle carbon intensity of a disaster-monitoring WSN?	**SO2:** Devise CALASH, a four-pillar lifecycle-aware routing protocol (CADR, CARE, SHDR, LSE) with formal carbon-budget guarantees via Lyapunov optimisation over a 6G physical layer.
Q3	Is a zero-trust decision shield able to detect compromised or hallucinating routing policies with zero false positives and enforce graduated responses within the 10 ms O-RAN real-time budget?	**SO3:** Propose CASTER-ZT, a GraphSAGE-based trust-risk evaluator coupled with a deterministic five-level graduated-decision shield, guaranteeing zero false positives within the 10 ms O-RAN near-RT RIC latency budget.

Methodological Approach

Framework: Design Science Research (DSR)

1. Problem identification (UNDRR, IPCC)
2. Objective definition (Q1, Q2, Q3)
3. Artifact design (GAPF, CALASH, CASTER-ZT)
4. Demonstration (simulation)
5. Evaluation (>23,800 runs)
6. Communication (3 articles)



Common Experimental Protocol

Simulation Environment

- Python + PyTorch, PyMARL2
- Heinzelman first-order energy model
- Real-world data: Intel Lab, UK Carbon Intensity API, USGS ShakeMap
- Seismic model calibrated on Turkey–Syria 2023

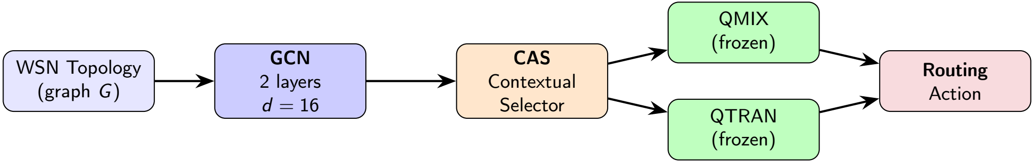
Statistical Rigor

- Article 1: 21,000 runs, 7 scenarios
- Article 2: 390 runs, 30 Monte Carlo seeds
- Article 3: 2,420 runs
- Total: $>$ 23,800 runs
- Statistical tests: 95% CI, Wilcoxon, bootstrap

GAPF: Graph-Attentive Policy Fusion

Key Idea

Fuse complementary frozen MARL expert policies (QMIX & QTRAN) via a lightweight GNN encoder, instead of training a heavier monolithic network.



- GCN: encodes topology into a compact embedding
- CAS: contextual selection of the optimal expert
- Monotonic Q·K Attention Network for multi-objective reward scalarization
- Tri-component: REINFORCE (GCN+CAS) + ES (Q·K attention) outer; MARL inner

GAPF: Why Fusion Works

QMIX / QTRAN Complementarity

- **QMIX**: $\partial Q_{\text{tot}} / \partial Q_i \geq 0 \Rightarrow$ tight coordination
- **QTRAN**: affine transforms $\Rightarrow$ non-monotonic interactions
- CAS selects expert by topological context

Monotonic Q·K Attention

- Multi-objective scalarization (energy, PDR, latency)
- Non-negative weights $\Rightarrow$ **Pareto monotonicity**
- Dominance $\Rightarrow$ higher scalar score
- Eliminates classical scalarization artifacts

"Expert Portfolio" Paradigm

Established in combinatorial optimization — applied 1$^{\text{st}}$ time to MARL routing in WSNs.

GAPF: Key Results

Metric	Baseline (QMIX)	GAPF	Gain
PDR (delivery ratio)	46–67%	$\geq$ **98%**	+43–111% $\uparrow$
Energy per packet	1.0×	0.29–0.50×	**2–3.5**× $\downarrow$
Trainable parameters	∼39K/agent	**1,473**	26× $\downarrow$
Training memory	3.2–13.8 GB	**951 MB**	3.4–14.5× $\downarrow$
Model size	—	**5.8 KB**	Cortex-M4 OK

✔ Contributions

- 1st MARL policy fusion framework via GNN for WSNs
- 26× fewer parameters than an RNN agent
- Deployable on microcontroller without compression

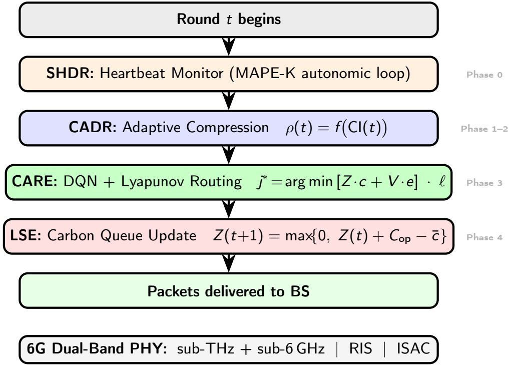
Validated on

- 7 scenarios (20–100 sensors)
- 21,000 test episodes
- Compared to QMIX, QTRAN, QPSOFL

CALASH: Carbon-Aware Lifecycle-Autonomous Self-Healing

Fundamental Insight

Embodied carbon dominates operational by $10^6:1$ in WSNs. Real lever: **maximize useful delivered packets** to amortize embodied carbon.



CALASH: The Four Pillars in Detail

CADR — Data Reduction

- Sensing ratio modulated by grid carbon intensity
- "Dirty" electricity: compress more
- "Clean" electricity: compress less

CARE — Carbon-Aware Routing

- Hybrid Lyapunov + DQN router
- Virtual carbon queue ⇒ drift-plus-penalty
- **CDL Pareto bound**: provable convergence

SHDR — Self-Healing

- Autonomic MAPE-K loop
- Detects dead nodes post-disaster
- Rebuilds topology automatically

LSE — Lifecycle Engine

- ISO 14040 compliant
- Manufacturing + operation + end-of-life
- LCI: gCO_2eq / useful packet

CALASH: Key Results

Metric	Baseline (LEACH)	CALASH	Gain
Network lifetime	931 rounds	**1,486 rounds**	+60%
PDR	—	**82.9%**	—
LCI (gCO_2eq/pkt)	13.42	**8.99**	−33%
Physical layer	classical	**6G dual-band**	sub-THz + RIS

✔ Contributions

- 1st LCI metric for WSNs (ISO 14040)
- 1st 4-pillar protocol with Lyapunov guarantees
- Insight: reducing transmitted volume > reducing energy/packet

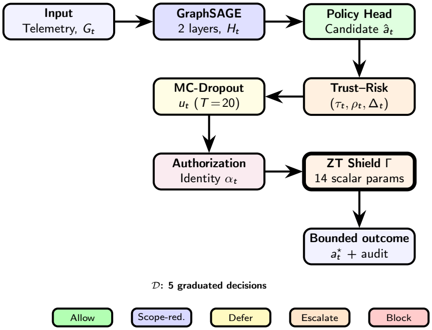
Validated on

- 390 runs, 30 Monte Carlo seeds
- Seismic model calibrated (Turkey 2023)
- UK Carbon Intensity API
- Compared to LEACH, HEED, EERP

CASTER-ZT: Zero-Trust Shielded Decision Control

Observation

AI in the control loop creates a **new attack surface**. Existing mechanisms (CPO, binary shielding) only offer *allow* or *block* — unsuitable for critical missions.



- NIST SP 800-207 compliant (zero-trust)
- Every actuation verified in real time
- 10 formal propositions proven
- Shield is **explainable** and **auditable**

CASTER-ZT: Formal Guarantees and Results

10 Formal Propositions

- Monotone conservatism
- Unauthorized-actuation exclusion
- Conformal coverage
- Graceful degradation
- Shield correctness & completeness

Deterministic Shield

14 scalar params, non-learnable — last defense if neural parts compromised.

Metric	CASTER-ZT	vs. baseline
Rogue detection	74–98%	+14–28 pts
False positives	**0%**	Full elim.
ω_{rec}	0.733	3.7–5.2× ↑
Latency/decision	**3.07 ms**	4.9× ↓
O-RAN budget	$\leq$ 10 ms	✔

Validated on

2,420 runs, 100 hex. cells. Compared to CPO, binary shield.

Protocol Stack Synergies

Layer 1 — GAPF Energy-efficient routing *PDR $\geq$ 98%, 2–3.5$\times$ energy $\downarrow$*

Efficient routing $\Rightarrow$ lower C_{oper}.

Layer 2 — CALASH Carbon sustainability *LCI −33%, lifetime +60%*

$\downarrow$ traffic (CADR) $\Rightarrow$ $\downarrow$ attack surface

Layer 3 — CASTER-ZT Zero-trust security *0% FP, 3.07 ms/decision*

Security protects efficiency gains

Mutual Reinforcement

The three layers reinforce each other: efficiency $\rightarrow$ sustainability $\rightarrow$ security. No trade-off.

Total Cost

$\sim$27K params $\approx$ 1 RNN agent, $\sim$103 KB (FP32) $\Rightarrow$ Cortex-M7 OK. Full stack $<$ one QMIX agent budget.

Overall Quantitative Summary

Metric	Best baseline	Baseline val.	Our result	Gain
PDR	QMIX	46–67%	$\geq$ **98%**	+43–111% ↑
Energy / packet	QMIX	1.0×	0.29–0.50×	2–3.5× ↓
Parameters	RNN agent	∼39K	**1,473**	26× ↓
Training memory	QMIX	3.2–13.8 GB	**951 MB**	3.4–14.5× ↓
Network lifetime	LEACH	931 rounds	**1,486 rounds**	+60%
LCI (CO_2/pkt)	LEACH	13.42	**8.99**	−33%
Rogue det. (0% FP)	Best 0-FP	0%	**74–98%**	+74–98 pts
False positives	Shield-Binary	48.8%	**0%**	Full elim.
Latency / decision	O-RAN budget	10 ms	**3.07 ms**	3.3× ↓
Criteria covered (/12)	Best existing	1	**12**	12× ↑
Total test runs				> **23,800**

TinyML Deployability

Framework	Params	Memory	Target	Compression
GAPF	1,473	5.8 KB	Cortex-M4	None needed
CALASH	$\sim$5K	19.5 KB	Cortex-M4	INT8 ($\rightarrow$4.9 KB)
CASTER-ZT	$\sim$20K	78 KB	Cortex-M7	Distill.+INT8 ($\rightarrow$20 KB)
Full stack	**$\sim$27K**	**103 KB**	**Cortex-M7**	Mixed quant.

Comparison

MCUNet deploys $\sim$1M parameters on Cortex-M7.
Our stack: $\sim$27K $\Rightarrow$ **38$\times$ lighter**.

Implication

Deployable on microcontrollers costing a few dollars $\Rightarrow$ accessible to developing countries (where 95% of deaths occur).

Six Original Scientific Contributions

1. **C1**: First **MARL policy fusion framework via GNN** for WSN routing
   GNN+CAS paradigm transferable to any multi-agent system
2. **C2**: **Monotonic Q·K Attention Network** for multi-objective scalarization
   Pareto monotonicity formally guaranteed — absent from existing methods
3. **C3**: First **Lifecycle Carbon Intensity (LCI)** metric for WSNs
   ISO 14040 compliant — adoptable for any IoT protocol
4. **C4**: First **4-pillar protocol with Lyapunov guarantees** for carbon budget
   Provable convergence to near-optimal PDR AND bounded carbon budget
5. **C5**: First **zero-trust decision control** framework for AI-native networks
   5 graduated decisions + 10 formal propositions + explainable shield
6. **C6**: Demonstration of **complementarity** between efficiency–sustainability–security
   Integrated protocol stack: all 3 objectives reinforce each other

Impact and SDG Alignment

For Industry

- GAPF (5.8 KB): no costly gateway
- +60% lifetime $\Rightarrow \downarrow$ maintenance
- LCI: sustainability reporting (CSRD/SEC)
- Auditable shield: IEC 62443

For Governments

- LCI: standardized IoT indicator
- Sendai indicators C-2, C-6
- Open access, no proprietary lock-in

SDG	Contribution
9	Resilient infrastructure (1,473 params)
11	Reduce losses (PDR $\geq$ 98%)
12	Sustainability (LCI ISO 14040)
13	Climate resilience (CADR)
16	Transparency (shield 0 FP)

For Society

- PDR $\geq$ 98%: reliable alerts
- $-$33% LCI: toward carbon neutrality
- SHDR: post-disaster coverage

Acknowledged Limitations

Methodological

1. **Simulation-based** — real data but no hardware
2. **Simplified disaster model** — Gaussian, no cascades
3. **Analytical 6G PHY** — no 3D ray tracing

Technical

4. **Unvalidated integration** — not tested as stack
5. **Scalability $>$ 200 nodes** — not characterized
6. **Adaptive adversaries** — static rogues only
7. **No field validation** — no user study

Future Research Directions

Short (1–2 yr)

- Unified stack integration
- Hardware testbed (Zolertia, Coral, Jetson)
- Extended disaster models

Medium (2–4 yr)

- Federated learning
- O-RAN digital twins
- Multi-disaster / multi-site
- Formal cert. (Coq/Isabelle)

Long (4+ yr)

- Non-terrestrial (LEO, HAPS)
- LLMs as meta-controllers
- Standardization (ISO, 3GPP)

Transferability: Precision agriculture | Smart cities | Connected health | Drone swarms

Conclusion

Summary

This thesis proposed a **complete protocol stack** for intelligent routing in 6G-integrated disaster-monitoring WSNs, built around three complementary frameworks:

GAPF
MARL fusion via GNN
1,473 params, PDR $\geq$ 98%
Energy 2–3.5$\times \downarrow$

CALASH
4 pillars + 6G PHY
LCI $-$33%, lifetime +60%
CDL Pareto bound

CASTER-ZT
5 graduated decisions
0% FP, 3.07 ms
10 formal propositions

Key Message

AI, designed with **parsimony** ($\sim$27K params), **environmental awareness** (LCI), and **systematic verification** (zero-trust), is a reliable tool for critical environments.

POLYTECHNIQUE MONTRÉAL

LARIM
LABORATOIRE DE RECHERCHE
EN RÉSEAUTIQUE ET INFORMATIQUE MOBILE

Thank You!

Questions?

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Backup Slides

Convergence and Theoretical Guarantees

GAPF

- QMIX: converges within the class of monotonic functions (IGM)
- QTRAN: larger space via affine factorization
- CAS: *variance reducer* (Wolpert's stacking)
- Inference: $O(|E| \cdot d)$ for GCN

CALASH

- Lyapunov drift-plus-penalty optimization
- $O(1/V)$ convergence to optimum
- $V = 100$: -33% LCI + acceptable latency
- CDL bound: *simultaneous* convergence PDR + carbon budget

CASTER-ZT

- 10 formal propositions
- Proved by logical construction
- Monotonicity, exclusion, coverage
- Shield is correct and complete

Gap Analysis: Coverage of 12 Criteria

Criterion	LEACH	QMIX	CPO	Our stack
Energy-efficient routing	✓	✓		✓
Topology awareness (GNN)				✓
MARL policy fusion				✓
Pareto-monotonic scalarization				✓
Lifecycle carbon metric				✓
Carbon-aware data reduction				✓
Carbon budget guarantees				✓
Post-disaster self-healing				✓
6G physical layer				✓
Rogue policy detection			✓	✓
Graduated decisions (5 levels)				✓
Formal security guarantees				✓
Total (/12)	**1**	**1**	**1**	**12**

Dec-POMDP Complexity and Positioning

Theoretical Context

- Optimal Dec-POMDP resolution: **NEXP-complete**
- CTDE (QMIX, QTRAN): structured approximation reducing $|\mathcal{A}|^n \rightarrow \sum_i |\mathcal{A}_i|$
- GAPF: adds expert selection as an additional level
- PDR $\geq$ 98% is not guaranteed *optimal* in the absolute sense, but constitutes a high-quality approximation

Positioning in the Value Decomposition Landscape

- QMIX $\rightarrow$ WQMIX $\rightarrow$ QPLEX $\rightarrow$ FOP: increasingly heavy architectures
- GAPF: **orthogonal** path — fusing lightweight experts via a topology-informed meta-controller
- "Portfolio" paradigm: well-established in combinatorial optimization, novel for MARL routing